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Aaron VanLandingham

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Overview

PhD candidate developing hybrid-electric propulsion systems for next-generation aircraft. Research expertise at the intersection of multidisciplinary design optimization, control system design, and hardware testing.

Education

The Pennsylvania State University
PhD in Aerospace Engineering, — *Expected Spring 2026*
MS in Aerospace Engineering, (GPA: 3.92/4.0) — *May 2024*

University of Southern California
BS in Aerospace Engineering, (GPA: 3.60/4.0) *cum laude* — *May 2017*

Work Experience

The Pennsylvania State University — *University Park, PA*
Graduate Research Assistant — *Aug 2021 – Present*

- Conceptual design and development of hybrid-electric rotorcraft powertrain concepts:
  - Vehicle design optimization of hybrid-electric rotorcraft
  - Design and fabrication of powertrain emulation facility for full-scale control system prototyping
  - Hybrid energy management control system development and implementation in subscale hardware
- Research funded by the Penn State Vertical Lift Research Center of Excellence (VLRCOE)

The Boeing Company — *Long Beach, CA*
Advanced Concepts - Configuration Design Engineer — *Aug 2017 – June 2021*

- Execute trade studies to size and operate hybrid-electric transport aircraft
- Led development of proprietary electric aircraft sizing tool (MATLAB)
  - Implemented modular propulsion tool for sizing and analysis of hybrid-electric propulsion architectures
  - Developed flexible mission definition tool for analysis of flight profiles and trajectory optimization

Projects

The Pennsylvania State University — *University Park, PA*

Hybrid-Electric Powertrain Testbed Design and Fabrication — *May 2022 – Present*

- Design and fabrication of 3 kW hardware-in-the-loop (HIL) powertrain testbed
  - Integrate motors, controllers, power supplies, and loads for closed-loop torque and speed control
  - Design and build safety enclosures, emergency stops, and startup procedures (Solidworks)
  - Develop data acquisition and testbed monitoring system (Speedgoat Analog and Digital IO, NI Labview)
- Validate scaling and control algorithm to preserve dynamic similtude at differing torques, speeds (Simulink)
- Implement T700 turboshaft engine, ECU, and HMU for real-time execution (MATLAB, Simulink)

Model-Based Energy Management Controller Development — *Aug 2024 – Present*

- Implement model predictive control (MPC) energy management controller to coordinate turbomachinery/electric machine control inputs (MATLAB, YALMIP, Gurobi)
  - Minimize energy consumption across mission while satisfying state and input constraints
  - Investigate impact of prediction horizon length on mission feasibility and objective
- Reduced order modeling of engines, motors, inverters, and batteries for model predictive control
- Swtiched linear turboshaft engine model captures variations in thermal efficiency across operating envelope

Hybrid-Electric Vehicle + Powertrain Design Optimization *Aug 2021 – Present*

- Developed optimization framework (GPkit) for evaluating benefits of hybrid-electric rotorcraft powertrains
  - Formulate electric machine, battery, and gearbox models as Geometric Programs (GPs)
  - GP implementation of NDARC vehicle sizing model and tabular rotor performance methdology for single main rotor, compound coaxial, and tiltrotor configurations
  - Modular powertrain architectures built-up from component models, facilitating topology optimization of gear stages, sizing torques, and speed ratios
- Identify benefit mechanisms of electrification for large rotorcraft:
  - Variable rotor speed optimizes blade loading coefficient to reduce power required
  - Battery at peak power conditions enables engine downsizing and reduced energy consumption
  - Magnitude of benefits depend on mission, objective, and vehicle size-class

Hydrogen-Fueled Transport Aircraft *Aug 2021 – March 2024*

- Developed optimization framework (GPkit) evaluating alternative fuels (SAF and LH2) for single-aisle and regional turboprop commercial transports
  - Lifecycle model captures emissions from fuel production and non-CO2 impacts such as contrails
  - Operating cost model developed from FAA data captures direct and indirect operating costs
- Results
  - LH2 tanks increase fuselage length, resulting in increased energy consumption compared to Jet-A/SAF
  - Assumed fuel characteristics drive differences between aircraft operating costs and emissions
  - When considering uncertainties in emissions from fuel production and the warming impact of contrails, LH2 and SAF aircraft have similar emissions and cost impacts relative to Jet-A aircraft

The Boeing Company *Long Beach, CA*

NASA Electrified Aircraft Propulsion (EAP) Study *Feb 2018 – Sept 2018*

- Designed 2035 Vision Vehicles for Part 25 regional jet and single-aisle commercial aircraft
  - Developed proprietary electric aircraft sizing tool to identify promising hybrid-electric architectures and sizing approaches that minimized operating costs
  - Coordinated technology assumptions for consistency with internal technology roadmaps
- Identified "minimal hybrid" sizing methodology for parallel-hybrid aircraft - engine sized to cruise, motor power with batteries supplements at top of climb condition
- Presented results to sponsors and key company stakeholders

Thin-Haul Commuter Study *Aug 2017 – January 2018*

- Design and analysis of series-hybrid aircraft for Part 23 thin-haul commuter market (e.g. Pilatus PC-12)
- Developed direct operating cost models for hybrid-electric and non-electric aircraft
- Hybrid-electric aircraft with reduced design range showed large reductions in operating costs compared to in-service aircraft
- Compared to an equivalently sized non-electric aircraft, results showed little to no benefit relative to conventional aircraft sized for the same range

Single Aisle Product Development Study *Oct 2019 – Feb 2020*

- Evaluation of parallel-hybrid architecture for future single-aisle aircraft (FAR Part 25)
- Calibrated performance calculations to higher-fidelity product development tools and sizing constraints
- Implemented "minimal hybrid" sizing approach and evaluated impact on sizing, performance, and economics
- Weight impact of electrification increased total operating cost despite reduced engine maintenance costs

Electric Aircraft Demonstrator Program *Oct 2018 – Feb 2020*

- Evaluate suitability of Part 23 donor aircraft for retrofit with hybrid-electric propulsion system
  - Calibrate electric aircraft sizing tool to published aircraft pilot operating handbook (POH) data
- Develop hybrid-electric propulsion architectures for various flight test campaign phases testing all-electric, parallel-hybrid, and series-hybrid propulsion architectures
- Develop Design Reference Missions (DRMs) and maintain performance targets for flight test campaign
  - Support program through System Requirements Review (SRR) and Preliminary Design Review (PDR)

Skills

Data Acquisition: NI Labview, Speedgoat FPGA Analog and Digital IO, Arduino
Programming: Python (SciPy, GPkit, Mosek, CVXOPT), MATLAB (YALMIP, Gurobi), LaTeX, Git
CAD Modeling: Solidworks, NX, OpenVSP

Awards

Eagle Scout

Publications

Subscale Emulation of Full-Scale Hybrid Electric Aircraft Powertrains June 2025
Aaron VanLandingham, DK Hall, Herschel C. Pangborn
2025 IEEE Transportation Electrification Conference + Electric Aircraft Technologies Symposium,

Hybrid-Electric Powertrain Design for Enhanced Vehicle Performance May 2025
Aaron VanLandingham, DK Hall, EC Smith, RC Bill
Vertical Flight Society 81st Annual Forum, 10.4050/F-0081-2025-369

Optimization of Electrified, Variable Speed Drivetrain Concepts for Enhanced Vehicle Performance May 2024
Aaron VanLandingham, DK Hall, EC Smith, RC Bill
Vertical Flight Society 80th Annual Forum, 10.4050/F-0080-2024-1315

Environmental and Economic Impact of Transport Aircraft Using Sustainable Aviation Fuel or Liquid-Hydrogen as Alternative Fuels May 2024
Aaron VanLandingham
The Pennsylvania State University, MS Thesis,

Conceptual Design Optimization of Liquid-Hydrogen-Fueled Transport Aircraft for Environmental and Economic Performance June 2023
Aaron VanLandingham, DK Hall
AIAA AVIATION 2023 Forum, 10.2514/6.2023-3228

An Integrated Hybrid-Electric Airplane, Propulsion and Power Sizing Method Jan 2022
RE Carter, BM Smith, SC Cleary, *Aaron VanLandingham*
AIAA SCITECH 2022 Forum, 10.2514/6.2022-1558

References

Dr. Edward C. Smith
Distinguished Professor, Aerospace Engineering
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Dr. David K. Hall
Assistant Professor - PhD Advisor
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